Purchase-to-pay review: where invoices wait, and what to fix first

To: Head of Purchase-to-Pay Operations · From: R. Karan · Data: 251,734 PO lines, 1,595,923 logged steps, 1,975 vendors, POs created in 2018 (BPI Challenge 2019 event log)

Bottom line

Most invoices that take longer than 60 days to pay are probably not a process failure: 68% of them sit in one spend area (Packaging), which pays at a median of 69 days — the signature of longer agreed terms. The delays the company does control are upstream: invoices wait a median 9.6 days to be booked, and goods receipts are posted late enough to hold payments up.

Findings and recommended actions

Finding	Evidence	Action	Scale
1. Check the terms before chasing "late" payments	42% of PO lines are paid after 60 days, but 57 vendors (3% of them) account for 80% of those lines, and Packaging alone accounts for 68%.	Pull the contract terms for the 57 vendors. Where terms are 60 days or less, this is real late payment; where they are longer, re-baseline the KPI so it stops flagging agreed behaviour.	57 contracts to check; 80% of the slow-paid lines explained or confirmed
2. Invoices wait to be booked	Median 9.6 days from the vendor's invoice date to booking; 42% wait over two weeks. Booking is 96% manual, done by 29 people (219,288 bookings), although batch jobs already book 9,472.	Pilot e-invoicing / OCR capture with the highest-volume vendors and track invoice-date-to-booking weekly, with a target median of 3 days.	219,288 manual bookings in the log; every day removed comes straight off the payment clock
3. Late goods receipts hold payments	When the invoice arrives before the goods receipt is posted, the line is paid after 60 days 7.7 percentage points more often — comparing within the same vendor (589 vendors). Blocked invoices spend 33% of their wait before the block is lifted (median 6.0 days).	Set a 2-working-day posting target for goods receipts and report it by site. GR posting is 73% manual across 246 people; batch already posts 85,034, so extend automatic receipt to the vendors that support it.	917,953 invoice-days spent blocked
4. POs raised after the invoice	3,621 lines (1.4%, €49.4M) were ordered after the vendor had already invoiced — a median 12 days later. Trading & End Products produces 40% of them; in Real Estate, 52% of lines are retroactive.	Enforce "no PO, no pay" in Trading & End Products and Real Estate first, and brief the buyers behind the 83 vendors that make up 80% of cases.	€49.4M of spend committed before approval
5. Staff for week 4 of the month, not the average week	Manual work in the fourth week of the month (days 22-28) runs 25% above the other weeks. A forecast that knows this misses by 11% four weeks ahead, against 17% for a flat 4-week average (6 backtest folds). The obvious suspect, month-end close, does not explain it: weeks holding a month end are only 6% busier.	Plan the accounts-payable and receiving rota from the week-of-month forecast. Find out what lands in week 4 (a payment-run or reporting cut-off is the likely candidate) before moving work out of it.	about 19,627 manual steps in a typical week

Checked and ruled out

• Rework does not slow payment. Lines with changes or cancellations are paid after 60 days 41.7% of the time, against 41.6% without — no difference worth acting on.
• A machine-learning model adds little to a simple rule. Ranking new invoices by the vendor's past record scores AUC 0.825; gradient boosting scores 0.842. Among the riskiest 10% of invoices, the model is right 100% of the time and the rule 97% — use the rule.

What this analysis cannot tell you

• The log has no payment terms. "Paid after 60 days" uses the EU Late Payment Directive's default ceiling as a yardstick, not as the company's contract; the sensitivity table shows the rate at 30, 45 and 90 days.
• The log does not link a payment to the invoice it settles, so each PO line is measured on its first invoice to its first clearance. Lines with many invoices (framework orders) are represented by that first one.
• Values are line values ("cumulative net worth"), not invoice amounts; where a line has several invoices, its value is split evenly between them.
• The week-of-month pattern was spotted by looking at the whole weekly series. The forecast learns its multipliers only from weeks before each origin, but the choice to try it was informed by the data, so its backtest error is somewhat optimistic.
• One company, one year (POs created in 2018), anonymised vendors and users. Findings describe this process, not purchase-to-pay in general.
• The statutory late-interest figure (€11.5M) is an upper bound that assumes 60-day terms everywhere. Given finding 1, it is not quoted as a cost.

Every number above is produced by one command (`python run_all.py`, seed 20261014) and rendered from its output; the SQL behind each one is in the project's `sql/` folder.